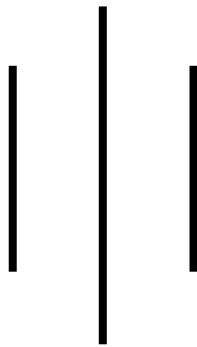


# INSTITUTE OF ENGINEERING

ADVANCED COLLEGE OF ENGINEERING AND MANAGEMENT  
KALANKI , KATHMANDU  
(AFFILIATED TO TRIBHUWAN UNIVERSITY)



## LAB REPORT

**SUBJECT:** Software Engineering  
**LAB NO:** 01

**SUBMITTED BY:**

**NAME:** Manu Sharan Kumar

**ROLLNO:** ACE080BCT037

**DATE:** 2083-02-17

**SUBMITTED TO:**

Department of Electronics  
and Computer Engineering

# Examination Management System

Software Requirement Specification

Software Engineering

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## Problem Statement

Conducting and managing examinations in an academic institution is a logistically demanding process. Administrators must simultaneously coordinate examination schedules, assign invigilators, allocate examination halls, manage student registrations, and process results — all while ensuring strict confidentiality and fairness. In institutions with hundreds or thousands of students, relying on manual or spreadsheet-based workflows makes the process error-prone, time-consuming, and difficult to audit. Common problems include scheduling conflicts, hall over-allocation, late result publication, and lack of a centralised record for student performance history.

## Proposed Solution

The **Examination Management System (EMS)** addresses these challenges by digitising and automating the end-to-end examination lifecycle.

Administrators can create and publish examination schedules, automatically detect conflicts, and assign invigilators based on availability. Students can register for examinations online, download admit cards, and view published results through a self-service portal. The system maintains a permanent, role-protected record of all marks and grade histories, and generates statistical performance reports at the individual, subject, and institutional level.

A dedicated notification module keeps all stakeholders informed of schedule changes, result releases, and pending tasks via in-app alerts and email.

## 1. Functional Requirements

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### R-1 User Registration and Role Assignment

#### *Description*

The system shall allow new users to create accounts with appropriate roles assigned during registration or by an administrator.

#### *Input*

Full name, institutional email, password, role (admin / faculty / student), department, roll number (students only).

#### *Process*

Validates all fields; checks for duplicate entries; hashes password; stores user record in the database; sends a verification email.

#### *Output*

Account created; verification email sent; user redirected to the login page.

### R-2 User Login and Session Management

#### *Description*

The system shall authenticate users and direct them to a role-appropriate dashboard.

#### *Input*

Institutional email and password.

#### *Process*

Matches credentials against stored records; verifies hashed password; generates a session token; logs login event with timestamp and IP address.

#### *Output*

User logged in and redirected to their role-specific dashboard (admin / faculty / student).

### R-3 Examination Schedule Creation

#### *Description*

The system shall allow administrators to create and publish examination schedules.

#### *Input*

Subject name, subject code, exam date, start time, duration, examination type (internal / final), eligible batch or semester.

#### *Process*

Validates all fields; checks for date and time conflicts across subjects and batches; stores schedule record; marks status as “Draft” until published.

#### *Output*

New examination entry appears in the schedule list; conflicts flagged before saving.

## **R-4 Hall Allocation**

### *Description*

The system shall allow administrators to assign examination halls to scheduled exams based on seating capacity.

### *Input*

Examination ID, hall name or ID, seating capacity required.

### *Process*

Checks hall availability for the selected date and time slot; verifies capacity against the number of registered students; records the allocation.

### *Output*

Hall assigned to the examination; allocation conflict shown if the hall is already booked or under-capacity.

## **R-5 Invigilator Assignment**

### *Description*

The system shall allow administrators to assign faculty members as invigilators for each examination session.

### *Input*

Examination ID, faculty member ID or name.

### *Process*

Checks faculty availability for the time slot; prevents double-assignment to overlapping sessions; stores the assignment and notifies the assigned faculty.

### *Output*

Invigilator listed under the examination; faculty member receives an in-app and email notification.

## **R-6 Student Examination Registration**

### *Description*

The system shall allow students to register for upcoming examinations within a defined registration window.

### *Input*

Student ID, examination session(s) to register for.

### *Process*

Validates that the student is eligible (enrolled in the relevant subject); checks that registration is within the open window; stores registration record.

### *Output*

Registration confirmed; examination listed under the student's upcoming exams.

## **R-7 Admit Card Generation**

### *Description*

The system shall generate and make available a downloadable admit card for each registered student.

### *Input*

Student ID, examination session ID.

### *Process*

Retrieves student registration details, hall allocation, and exam schedule; compiles admit card data; generates a PDF document.

### *Output*

Admit card PDF available for download from the student dashboard, containing hall number, seat number, exam timetable, and student details.

## **R-8 Mark Entry and Validation**

### *Description*

The system shall allow authorised faculty to enter and submit marks for students after examinations.

### *Input*

Subject ID, student roll number, marks obtained, full marks, pass marks.

### *Process*

Validates that entered marks do not exceed full marks; flags entries below the pass threshold; stores records with the submitting faculty's ID and timestamp; marks status as "Pending Approval".

### *Output*

Marks recorded in the system; flagged entries highlighted for admin review before publication.

## **R-9 Result Publication**

### *Description*

The system shall allow administrators to review, approve, and publish examination results.

### *Input*

Examination ID, admin approval action.

### *Process*

Admin reviews pending mark entries; approves or returns for correction; on approval, results status changes to "Published" and becomes visible to students.

### *Output*

Results accessible on student dashboards; notification sent to all registered students that results are available.

## **R-10 Grade Calculation**

### *Description*

The system shall automatically calculate letter grades and grade points based on the institution's grading policy.

### *Input*

Marks obtained, full marks, configured grading scale.

### *Process*

Computes percentage; maps to the configured grade boundary table; calculates grade point; computes GPA/CGPA where applicable.

### *Output*

Grade and grade point displayed alongside marks on the result view; CGPA updated on the student's academic record.

## **R-11 Notification and Alert System**

### *Description*

The system shall send timely notifications to relevant users about examination-related events.

### *Input*

Trigger event (schedule published, result released, registration deadline approaching, invigilator assigned).

### *Process*

On each trigger event, identifies the affected user group; generates notification message; dispatches via in-app notification and optionally via email.

### *Output*

Notification appears in the user's notification centre; email delivered to registered address.

## **R-12 Performance Report Generation**

### *Description*

The system shall generate statistical reports on student performance at subject and batch level.

### *Input*

Report type (subject-wise / batch-wise / individual), examination session or date range.

### *Process*

Queries published result records within the selected scope; calculates pass rate, average marks, highest and lowest marks, and grade distribution; formats data into a report.

### *Output*

Report displayed on screen with option to export as PDF or CSV.

## **R-13 Academic Transcript Generation**

### *Description*

The system shall generate an official academic transcript for a student covering all completed examinations.

### *Input*

Student ID, optional date range or semester filter.

### *Process*

Retrieves all published result records for the student; compiles subject-wise marks, grades, and CGPA; formats into a printable transcript layout.

### *Output*

Transcript PDF generated and available for download from the student dashboard; admin can generate and print an official copy.

## **R-14 Examination Schedule Publication and Visibility**

### *Description*

The system shall allow admins to publish finalised schedules so they become visible to students and faculty.

### *Input*

Examination schedule ID, admin publish action.

### *Process*

Changes schedule status from “Draft” to “Published”; triggers notification to all affected students and invigilators.

### *Output*

Schedule visible on student and faculty dashboards; notifications dispatched.

## **R-15 Role-Based Access Control**

### *Description*

The system shall restrict access to features and data based on the authenticated user’s role.

### *Input*

User role (admin / faculty / student) assigned at registration.

### *Process*

On every request, the system verifies the user’s role against a permissions table; unauthorised actions are blocked at the API level with an appropriate error.

### *Output*

Admin has full system access; faculty can enter marks and view assigned invigilations; students can only access their own registration, admit card, and results.

## 2. Non-Functional Requirements

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### NR-1 Page Load Performance

#### *Description*

The system shall load the main dashboard within an acceptable time to ensure a smooth user experience.

#### *Input*

User navigates to the dashboard after login.

#### *Process*

Server fetches schedule and result data, compiles summary metrics, and renders the dashboard view.

#### *Output*

Dashboard fully loaded and interactive within 3 seconds under normal network conditions.

### NR-2 System Scalability

#### *Description*

The system shall support concurrent usage by a large number of students without performance degradation, particularly during result publication.

#### *Input*

Multiple simultaneous user sessions accessing the result portal.

#### *Process*

Stateless API design and database query optimisation (indexing on student ID, subject ID, and examination ID) ensure efficient data retrieval under load.

#### *Output*

System remains responsive for at least 500 concurrent users with no timeout errors.

### NR-3 Data Integrity and Accuracy

#### *Description*

The system shall ensure that marks and grade records are accurate and cannot be unintentionally overwritten.

#### *Input*

Mark entry or update request from faculty.

#### *Process*

Input validated before saving; database transaction locks prevent race conditions on simultaneous updates; all changes logged with user ID and timestamp.

#### *Output*

Records displayed are always consistent with the most recent valid, approved entry; audit trail maintained for every change.

### NR-4 Role-Based Data Privacy

#### *Description*

The system shall ensure users can only access data they are authorised to view.

#### *Input*

A student or faculty user attempts to access another student's results or an admin-only module.

#### *Process*

Role check performed on every API request; unauthorised access blocked; no restricted data included in the response payload.

#### *Output*

Unauthorised user receives a permission error; no sensitive data exposed.

### **NR-5 Password Security**

#### *Description*

The system shall store user passwords securely to prevent unauthorised access in the event of a data breach.

#### *Input*

User password entered during registration or password change.

#### *Process*

Password hashed using bcrypt or a similarly strong algorithm before storage; plain-text password is never persisted.

#### *Output*

Only the hashed password is stored; the original password is unrecoverable from stored data.

## **3. Additional Constraints**

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### **C-1 Minimum Data for Grade Calculation**

The grade calculation module requires that the grading boundary table be fully configured by an administrator before any results can be published. Results cannot be approved without a valid, active grading policy in place.

### **C-2 Internet Dependency**

The web portal requires a stable internet connection for all operations. There is no offline mode; invigilators and students must have network access during examination sessions where the system is used for attendance or real-time updates.

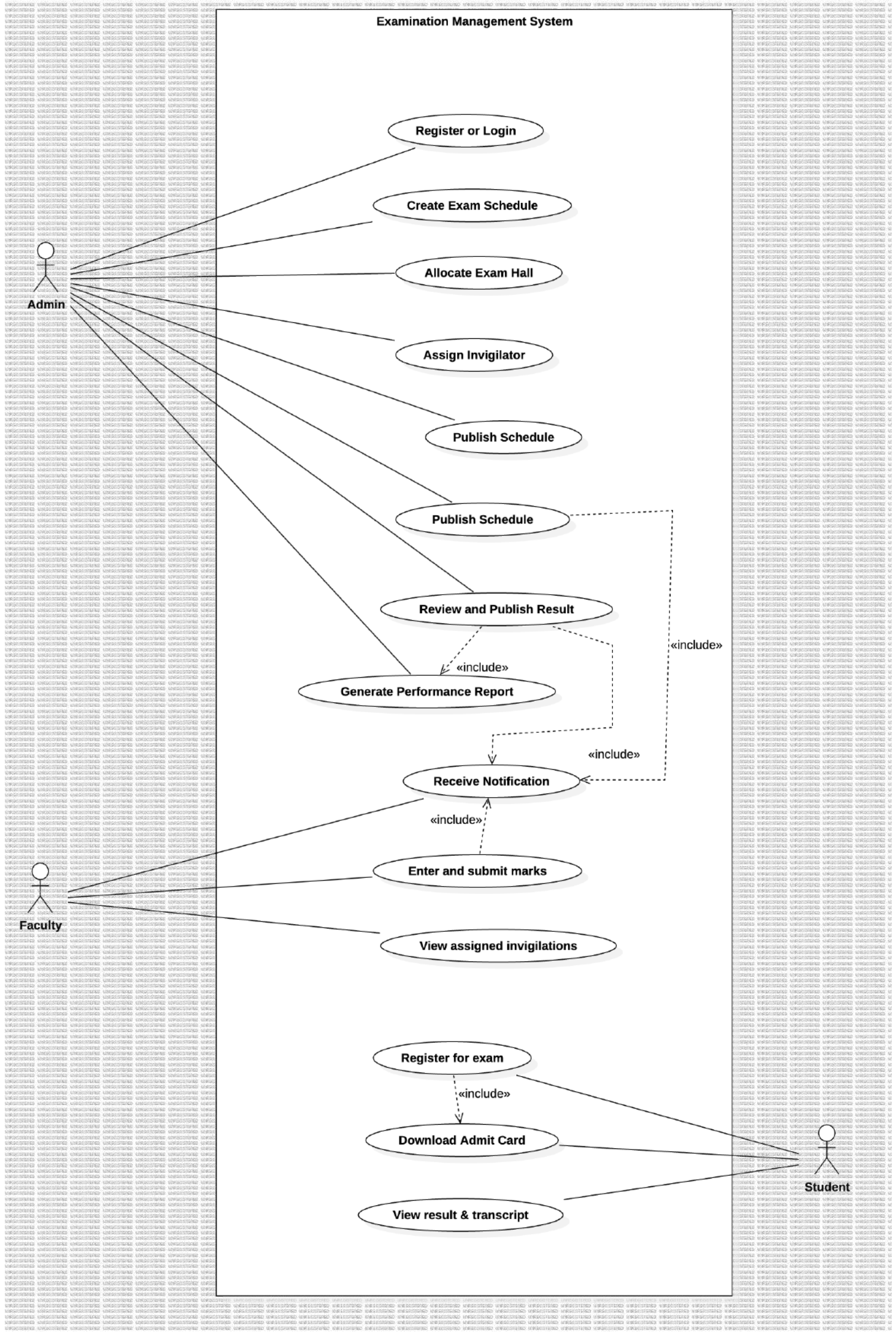
### **C-3 Academic Scope**

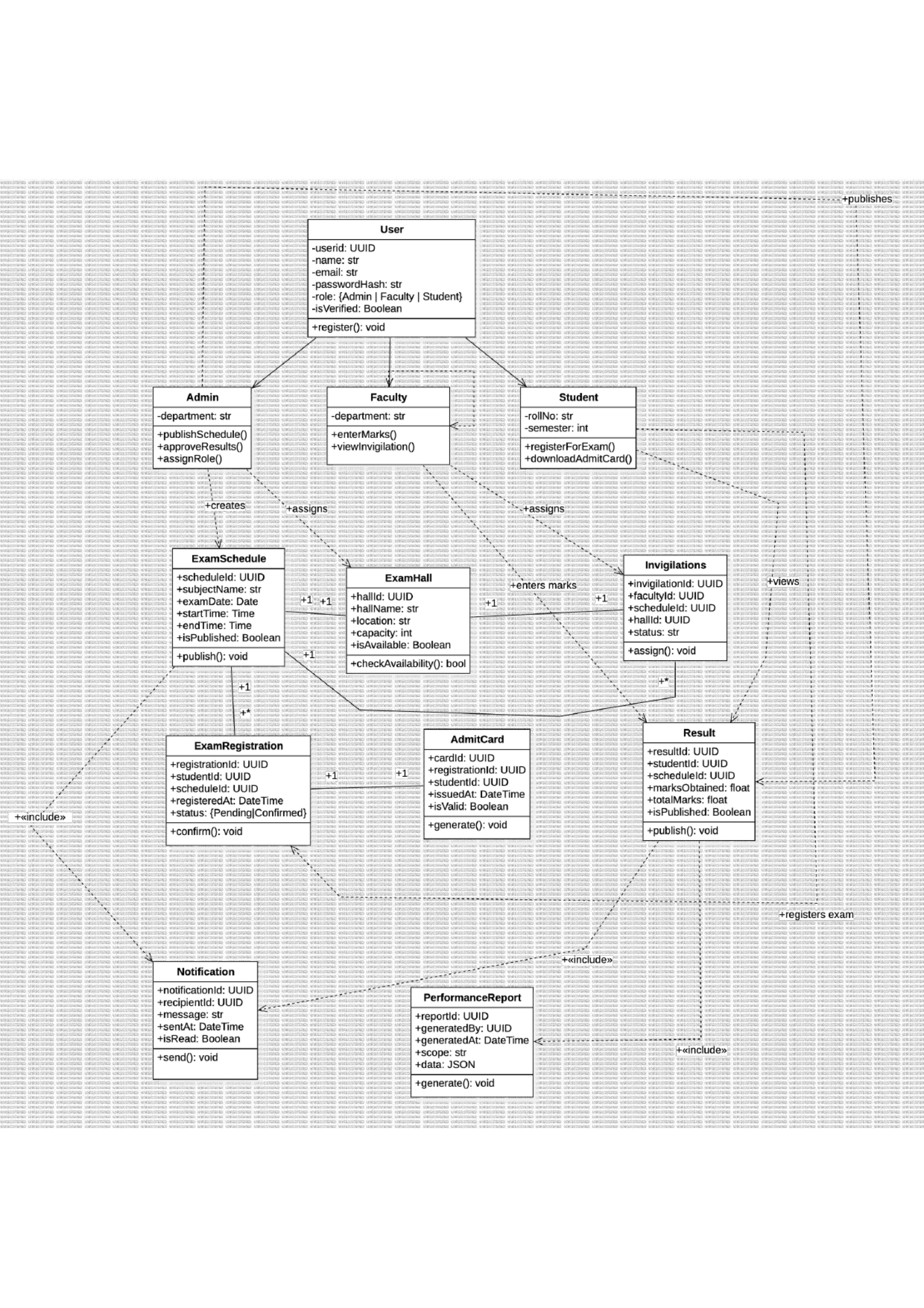
The system manages examination scheduling, marks, and results only. It does not function as a full student information system (SIS) or learning management system (LMS). Integration with third-party SIS platforms is a planned future enhancement.

### **C-4 Single Institution Scope**

Version 1.0 supports a single institution and campus. Multi-campus or multi-institution support is planned for a future release.

# Examination Management System





# 1 System Workflow

The Examination Management System operates through a role-driven workflow covering account access, administrative scheduling, student self-service, and faculty mark entry. Each stage is illustrated below with the corresponding interface.

## 1.1 Login and Registration

The login page is the entry point of the EMS. Users authenticate with their institutional email and password (R-2); new users can register and are assigned a role of Admin, Faculty, or Student (R-1). On successful login, the system redirects the user to a dashboard scoped to their role, as enforced by role-based access control (R-15).

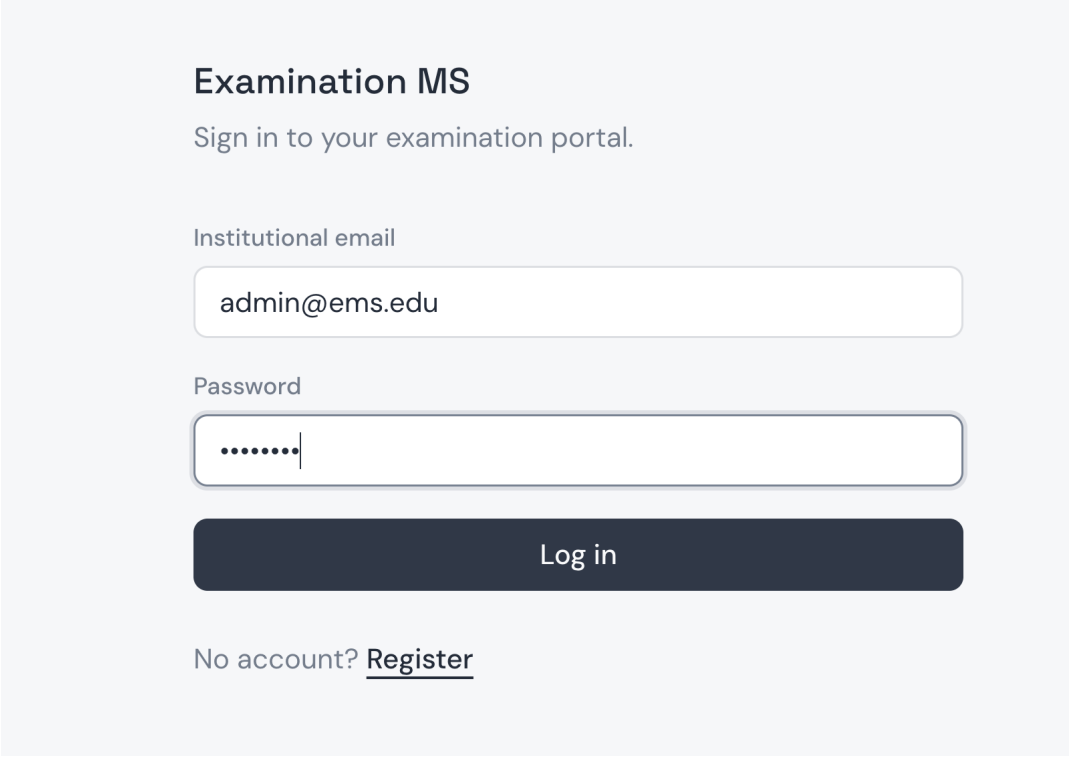
The image shows a login and registration interface for the 'Examination MS'. At the top, the title 'Examination MS' is displayed in a bold, dark font. Below it, a subtitle reads 'Sign in to your examination portal.' in a smaller, lighter font. The form consists of two input fields: 'Institutional email' and 'Password'. The email field contains the text 'admin@ems.edu'. The password field is masked with dots. Below these fields is a dark blue button with the text 'Log in' in white. At the bottom of the form, there is a link that says 'No account? Register'.

Figure 1: EMS login and registration screen

## 1.2 Admin Dashboard — Schedule, Hall, and Invigilator Management

The Admin dashboard is the central control panel of the system. From here, administrators create examination schedules (R-3), allocate examination halls based on seating capacity (R-4), and assign invigilators while the system automatically prevents double-booking of both halls and faculty (R-5). Conflicts are flagged at the point of entry rather than after publication, and a schedule remains in *Draft* status until the administrator explicitly publishes it (R-14).

Examination MS

Admin Portal

Overview

Exam schedules

Examination halls

Results review

Grading policy

Performance reports

Users

Log out

Exam schedules

Create, publish and staff examination sessions

Subject name

Mathematics

Subject code

ENCT037

Eligible batch

2080-BCT

Exam date

25/08/2026

Start time

14:00

Duration (minutes)

180

Exam type

Internal

Create schedule

All schedules

| SUBJECT                       | DATE / TIME           | BATCH    | TYPE  | STATUS    |        |
|-------------------------------|-----------------------|----------|-------|-----------|--------|
| Software Engineering<br>SE401 | 2026-08-18 · 10:00:00 | 2080-BCT | Final | Published | Manage |

Figure 2: Examination schedule creation and conflict checking

Examination MS

Admin Portal

Overview

Exam schedules

Examination halls

Results review

Grading policy

Performance reports

Users

Log out

Examination halls

Manage hall capacity and allocations

Add examination hall

Hall name

RIU Hall

Location

Fourth Floor

Seating capacity

60

Add hall

Registered halls

| NAME   | LOCATION              | CAPACITY |
|--------|-----------------------|----------|
| Hall A | Block 1, Ground Floor | 60       |

Figure 3: Hall allocation with capacity and availability checks

### 1.3 Student Registration and Admit Card

Once a schedule is published, eligible students (matched by department / batch) can register for the examination within the open registration window (R-6). After registration is confirmed, the student can generate and download a PDF admit card containing their assigned hall, seat, and exam timetable (R-7).

All schedules

| SUBJECT                       | DATE / TIME           | BATCH    | TYPE  | STATUS    |       |
|-------------------------------|-----------------------|----------|-------|-----------|-------|
| Software Engineering<br>SE401 | 2026-08-18 · 10:00:00 | 2080-BCT | Final | Published | Close |

HALL ALLOCATION

Hall A — 60 seats

Hall A (cap 60)

45

Allocate

INVIGILATORS

Dr. Faculty Member

Dr. Faculty Member

Assign

REGISTRATIONS (1)

Seat 1 — a89b379e

Figure 4: Invigilator assignment with double-booking prevention

Examination MS

Student Portal

Exams & admit cards

Results & transcript

Exams & admit cards

Register for upcoming examinations and download admit cards

Manu Sharan Kumar

Registration confirmed. Your admit card is now available.

Available examinations — 2080-BCT

| SUBJECT                       | DATE / TIME           | TYPE  |                     |
|-------------------------------|-----------------------|-------|---------------------|
| Software Engineering<br>SE401 | 2026-08-18 · 10:00:00 | Final | Download admit card |

Figure 5: Student examination registration screen

## **EXAMINATION ADMIT CARD**

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Student Name: Manu Sharan Kumar

Roll Number: ACE080BCT037

Department / Batch: 2080-BCT

Subject: Software Engineering (SE401)

Exam Type: ExamType.final

Exam Date: 2026-08-18

Start Time: 10:00:00

Duration: 180 minutes

Examination Hall: Hall A

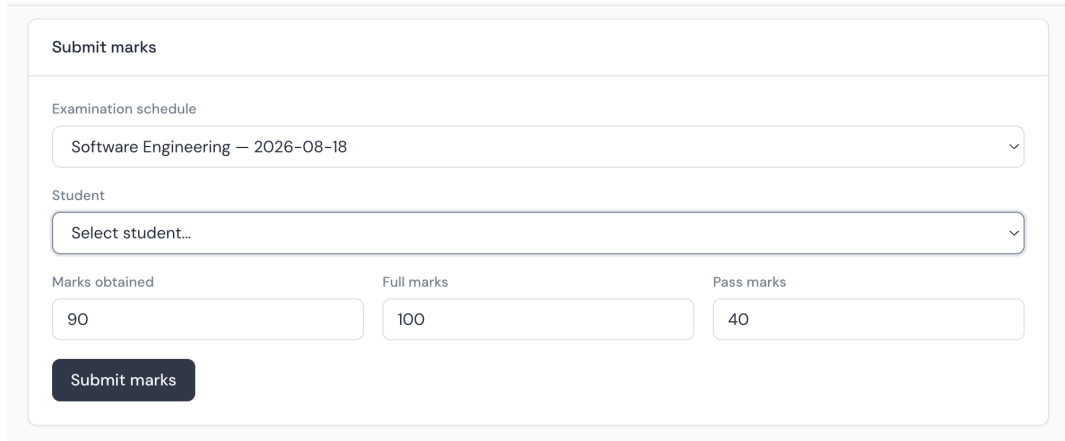
Seat Number: 2

Registration ID: 9155fe12-f595-46de-9c9d-8a431f72bd56

Figure 6: Downloadable admit card showing hall, seat, and timetable

## 1.4 Faculty Mark Entry

After an examination is conducted, authorised faculty enter marks for students under their subject. The system validates entries against the configured full marks and flags submissions below the pass threshold before they are sent for administrative review (R-8).

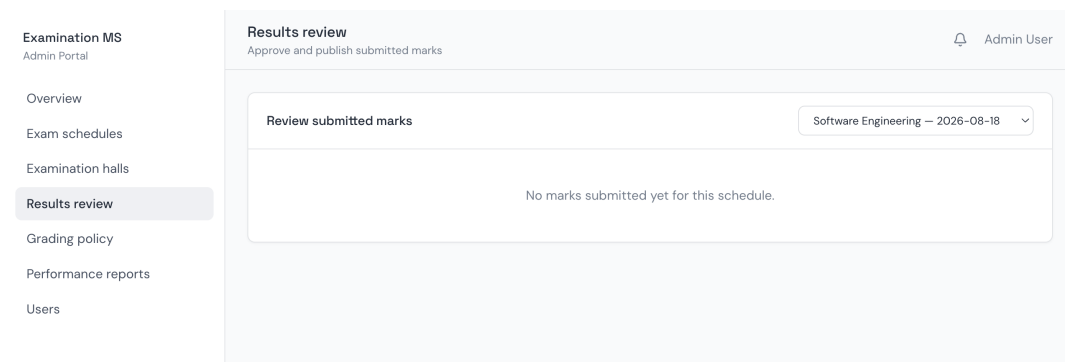


The screenshot shows a 'Submit marks' form. At the top, there's a section for 'Examination schedule' with a dropdown menu showing 'Software Engineering – 2026-08-18'. Below that is a 'Student' section with a dropdown menu labeled 'Select student...'. Underneath are three input fields: 'Marks obtained' with the value '90', 'Full marks' with the value '100', and 'Pass marks' with the value '40'. At the bottom left of the form is a dark blue button labeled 'Submit marks'.

Figure 7: Mark entry with full-marks validation

## 1.5 Result Review, Publication, and Grade Calculation

Administrators review pending mark submissions and either approve or return them for correction (R-9). On approval, the system automatically computes the letter grade and grade point for each result using the configured grading scale (R-10), provided a valid grading policy has already been set up, per constraint C-1. Once published, results become visible on the student's dashboard, and a notification is dispatched to all registered students (R-11).



The screenshot shows the 'Results review' interface. On the left is a sidebar menu for 'Examination MS Admin Portal' with options: Overview, Exam schedules, Examination halls, Results review (highlighted), Grading policy, Performance reports, and Users. The main content area is titled 'Results review' with a subtitle 'Approve and publish submitted marks' and a user indicator 'Admin User'. Inside the main area, there's a section 'Review submitted marks' with a dropdown menu showing 'Software Engineering – 2026-08-18'. Below this, a message states 'No marks submitted yet for this schedule.'

Figure 8: Result approval workflow and automatic grade calculation

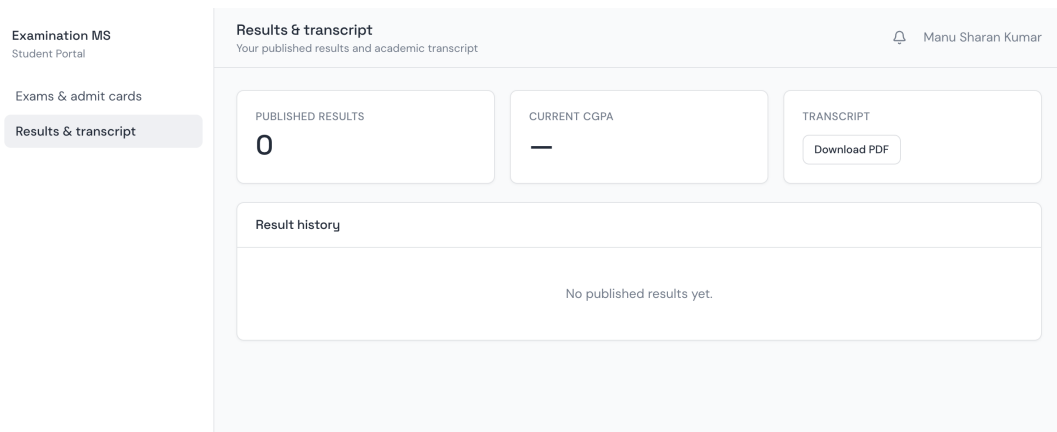


Figure 9: Student view of a published result with computed grade

## 1.6 Notifications

Throughout the workflow, the notification module keeps all stakeholders informed. In-app alerts (and optionally email) are dispatched automatically whenever a schedule is published, an invigilator is assigned, or results are released (R-11).

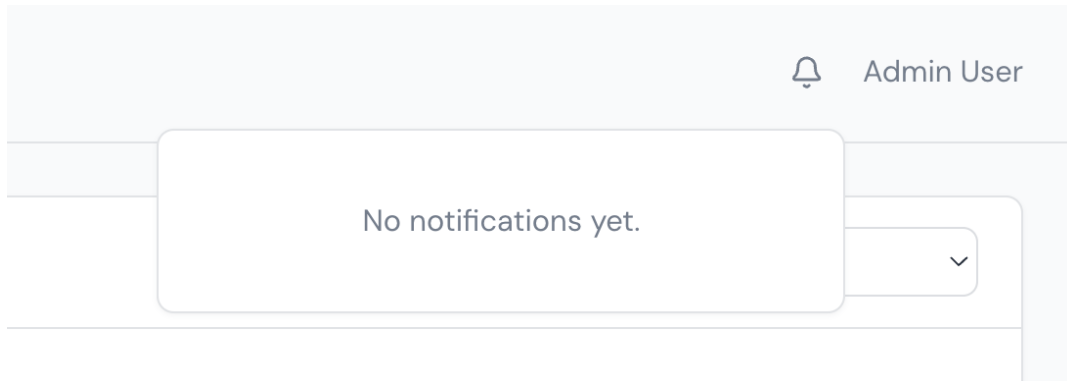


Figure 10: In-app notifications for schedule and result events

## 1.7 Performance Reports and Academic Transcript

Administrators can generate subject-wise or batch-wise performance reports — including pass rate, average marks, and grade distribution — exportable as PDF or CSV (R-12). Students can generate an official academic transcript summarising all published results with their cumulative CGPA (R-13).

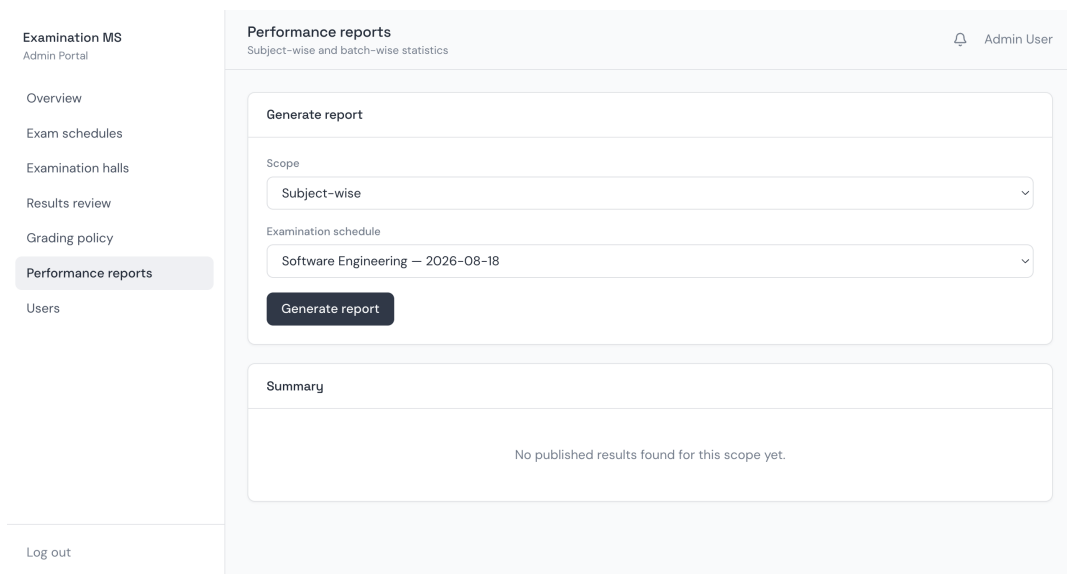


Figure 11: Subject-wise / batch-wise performance report

## OFFICIAL ACADEMIC TRANSCRIPT

Student Name: Manu Sharan Kumar  
Roll Number: ACE080BCT037  
Department / Batch: 2080-BCT

| Subject                         | Code | Marks Obtained | Full Marks | Grade | Grade Point |
|---------------------------------|------|----------------|------------|-------|-------------|
| No published results on record. |      |                |            |       |             |

**Cumulative GPA (CGPA): N/A**

Figure 12: Generated academic transcript with CGPA

## 2 Source Code Repository

The complete project source code, including the FastAPI backend and the React/TypeScript (TanStack Start) frontend, is available on GitHub:

[https://github.com/manusharansah/Exam\\_Management\\_System.git](https://github.com/manusharansah/Exam_Management_System.git)